

PRANAV N J

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Education

Siddaganga Institute of Technology , BE in Information Science	Sept 2023 – July 2027
• CGPA: 9.38/10	
Excellent PU College , PCMB	May 2023
• 12th : 94.5%	
JVS Primary and High School ,	May 2021
• 10th : 95.64%	

Skills

Languages: C++, C, Javascript

CS Fundamentals: Data Structures, DBMS, OOPs, OS

Web Development: HTML, CSS, JavaScript, Reactjs, Nodejs, Express.js, PostgreSQL, MongoDB

Tools and Technologies: Git, Github, VS Code

Soft Skills: Communication, Teamwork, Public Speaking, Logical Reasoning

Leadership Skills: National Cadets Corps, India

Projects

Phishing Detection System

- Architected enterprise-grade platform using FastAPI microservices, React, and Web Extension (Manifest V3), achieving 93.89% accuracy while reducing prediction latency to 29.74ms through data leakage prevention
- Eliminated HTTPS protocol bias by increasing training samples 100% (27,679 to 55,296 URLs) through dataset augmentation, reducing false positive rate to 4.79% and preventing production failure
- Developed Random Forest classifier with 59-feature extraction, improving accuracy by 4.4%, precision by 7.1%, and achieving 98.3% AUC-ROC through GridSearchCV tuning and StratifiedKFold 5-fold cross-validation
- Architected production-ready microservices with RESTful API, MongoDB, CORS security, and rate limiting (100 req/15min), reducing response time to sub-30ms while evaluating 4 ML algorithms for optimal selection
- Tools Used: Python, FastAPI, Scikit-learn, Pandas, React, Express.js, Node.js, REST API, MongoDB

Suraksha

- AI-powered women's safety platform built in TypeScript (React + Node + PostgreSQL) delivering sub-2s multi-channel emergency alerts, AI voice distress detection, and encrypted real-time coordination for families.
- Engineered AI emergency voice distress detection with a 3-stage pipeline (AssemblyAI → CoreNLP → Llama 2), reaching 96% accuracy and cutting false positives by 35% via weighted keywords plus rule-based fallback.
- Optimized 24/7 real-time location tracking with 15s GPS WebSocket beacons and WebRTC live video streaming, sustaining 97.8% multi-channel delivery across 37+ active emergency sessions.
- Tools Used: TypeScript, React, Node.js, Express.js, PostgreSQL(Drizzle ORM), REST API, WebSocket, WebRTC, AssemblyAI, CoreNLP, Llama 2, JWT + AES-256-GCM.

Achievements

Technical Competitions: Selected participant for the Smart India Hackathon 2025, a premier national-level hackathon by the Government of India; Participant at HackNite'25 organised by MIT Bengaluru; Participant at Dev Heat Hackathon hosted by IIIT Surat.

Leadership & Service: Leveraged disciplined leadership as a Certified National Cadet Corps (NCC) Cadet and Event Management Team Member for the Institution's Innovation Council (IIC-SET) to drive organizational excellence.